THE HONG KONG POLYTECHNIC UNIVERSITY
香港理工大學

COMP5523
Group Project Introduction

Feb 24, 2026

Opening Minds • Shaping the Future
啟迪思維 • 成就未來

00 Outline

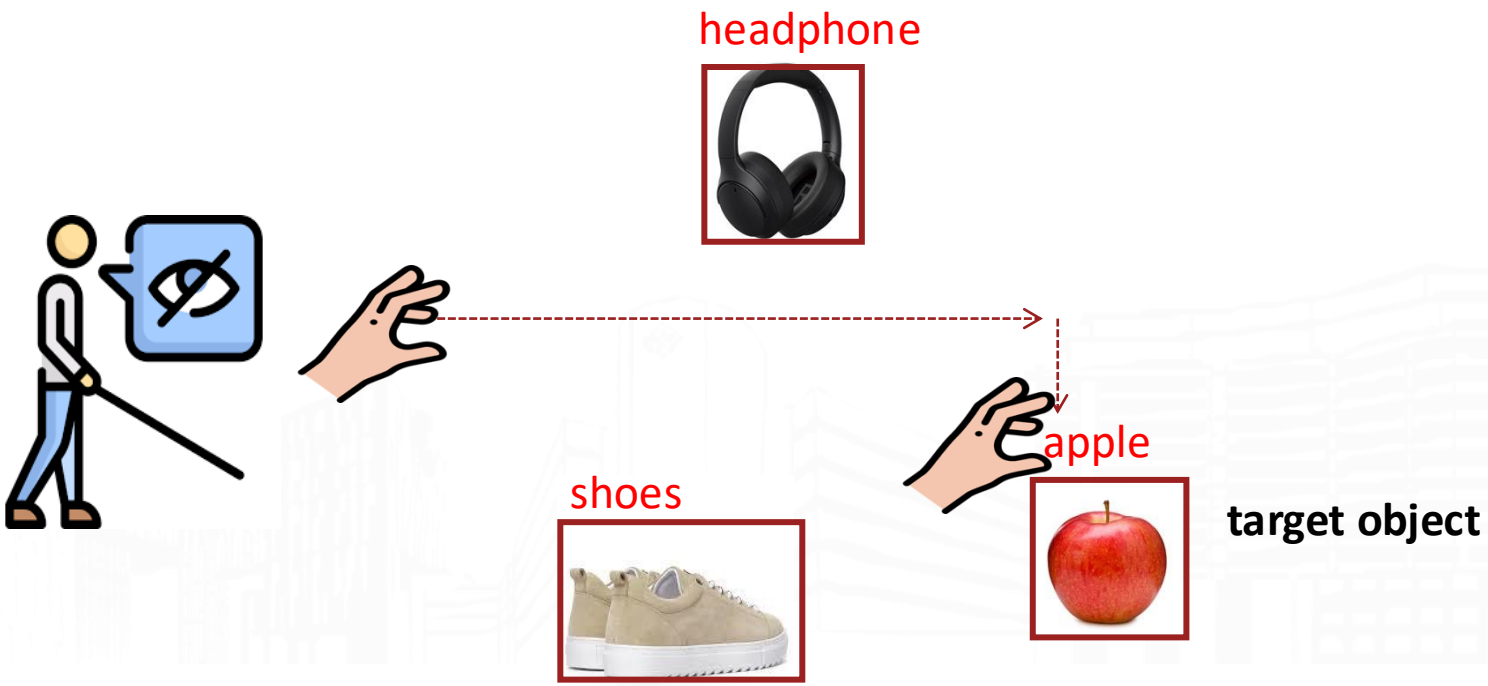
- Group Project Specification
- Group Presentation
- Feedback System

01 Group Project Specification

Topic: Interactive Vision-Assisted Object Grasping for the Visually Impaired

Develop a system to help visually impaired people safely and efficiently pick up a *specific target object* in daily life.

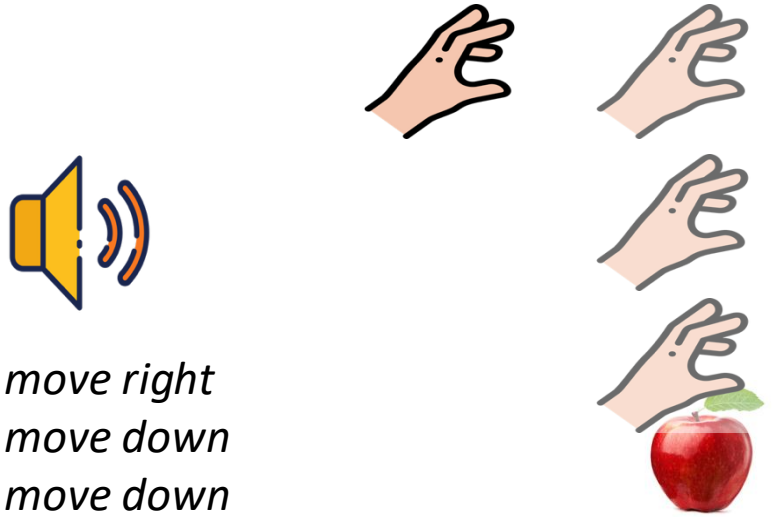
The object selection is not limited, choose whatever common objects in daily life



01 Group Project Specification

Core Idea & Relevant techniques

- Track the user's hand position
  - *object detection*
  - *hand tracking*
- Track the target object's position
  - *object detection*
- Spatial understanding
  - *Spatial relative distance measuring (hand <--> object)*
- Real-time audio guidance
  - *ASR, TTS*



01 Group Project Specification

Project Steps

- **Data preparation**: datasets for objects, hand pose, reaching/grasping videos
- **Algorithm design**: compare detection + tracking; spatial relation understanding; define guidance strategy
- **System implementation**: real-time pipeline + user-friendly audio interface
- **Evaluation**: success rate, guidance accuracy, time-to-grasp, latency, failures

01 Group Project Specification

Example Video

01 Group Project Specification

Difference between Last Semester's Project and This Semester's

- Last semester: more general; focused on general scene understanding, e.g.,
  - **Obstacle detection**: Detect objects in front of visually impaired users and alert them when obstacles are found
  - **OCR (Optical Character Recognition)**: Read and convey text content from images to visually impaired users
  - **Scene Description**: provide comprehensive descriptions of the surrounding environment to visually impaired users

- This semester: more task-specific; guide users to pick up objects through real-time audio guidance (e.g., "move left", "closer", "open hand").

02 Group Project Presentation

Presentation Date

- The group project presentation will happen on 📅 **April 14th**
- Each group has **7 minutes** to showcase your project, and **there is no a Q&A session.**

02 Group Project Presentation

Presentation Order

Will be released before presentation

02 Group Project Presentation

Presentation Process

- Presenter(s) join the zoom meeting in advance; share screen when start presenting; strictly control the time within 7mins
  Please install Zoom, sign in with your PolyU account, grant screen-sharing and voice permission in advance.

- The professor and TAs will evaluate and score during the presentation.

- Students from other groups will provide feedback via an online feedback system.
  Students who submit constructive feedback that is endorsed by the presenting group will earn maximum 1 bonus point , which will count toward the final score (0-100 scale)

- After finishing the presentation, presenting group should review the feedback online and endorse helpful comments.

02 Group Project Presentation

Tips for a successful presentation

Given the limited time (7 minutes):

- Please ensure your equipment and environment are properly set up.
- Please don't let your laptop die; 🔋 fully charge it before presenting.
- Please configure Zoom in advance
  - *install Zoom*
  - *sign in with your PolyU account*
  - *grant screen-sharing and voice permission*
- Practice beforehand and keep your presentation within 7 minutes.

02 Group Project Presentation

Scoring Rubics

1) Appropriateness.

- Task settings, challenges, methodologies, and system functionality are highly appropriate and relevant. (3%)
- Task settings, challenges, methodologies, and system functionality are mostly appropriate and relevant. (2.5%)
- Task settings, challenges, methodologies, and system functionality demonstrate some level of appropriateness. (2%)
- Task settings, challenges, methodologies, and system functionality are inadequate. (1.5%)
- Task settings, challenges, methodologies, and system functionality are inappropriate. (1%)

02 Group Project Presentation

Scoring Rubics

2) Soundness.

- The project demonstrates a comprehensive and well-organized development process with clear and logical explanations. (3%)
- The project demonstrates a mostly comprehensive and well-organized development process with mostly clear and logical explanations. (2.5%)
- The project demonstrates some level of organization and logic in the development process, but with some unclear explanations. (2%)
- The project demonstrates limited organization and logic in the development process and provides inadequate explanations. (1.5%)
- The project is disorganized and lacks a clear or logical development process with unclear or illogical explanations. (1%)

02 Group Project Presentation

Scoring Rubics

3) Excitement.

- The project presents innovative and engaging ideas that consistently capture the attention of the audience. (3%)
- The project presents engaging ideas that mostly capture the attention of the audience, but may lack consistency. (2.5%)
- The project presents some level of engagement, but with areas of improvement and inconsistent audience attention. (2%)
- The project lacks innovation and fails to capture the audience's excitement consistently. (1.5%)
- The project lacks excitement or fails to engage the audience. (1%)

02 Group Project Presentation

Scoring Rubics

4) Presentation.

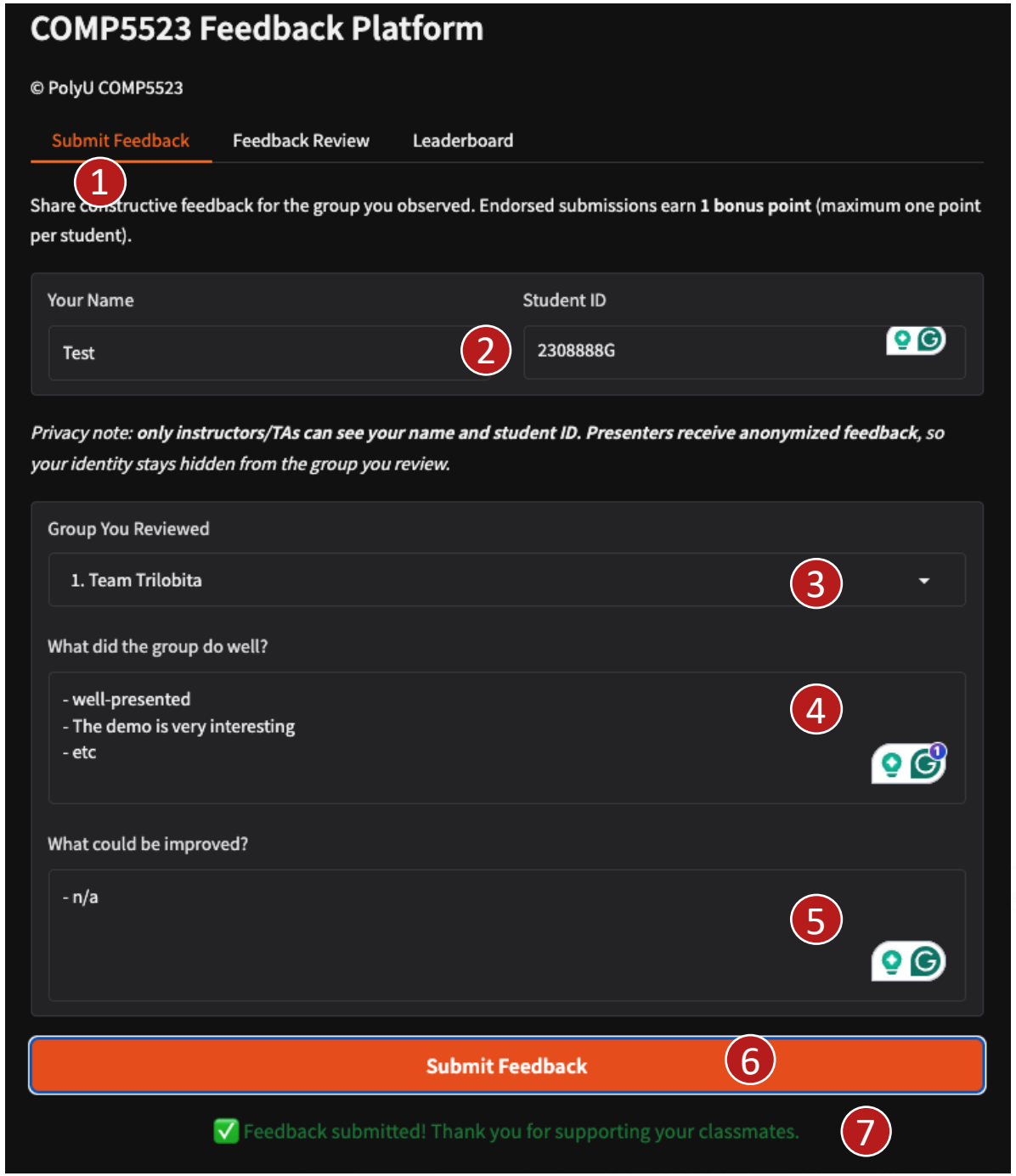
- The presentation is highly polished and professional, with excellent delivery and effective use of visual aids. (3%)
- The presentation is mostly polished and professional, with good delivery and adequate use of visual aids. (2.5%)
- The presentation has some areas of improvement in polish and professionalism, and delivery may lack consistency. (2%)
- The presentation lacks polish and professionalism, and delivery is weak. Visual aids are ineffective. (1.5%)
- The presentation is poorly executed, making it difficult to follow, with ineffective use of visual aids. (1%)

02 Group Project Presentation

Scoring Rubics

5) Writting.

- The project report is well-written, with clear and concise explanations of ideas and proper use of grammar and formatting. (3%)
- The project report is mostly well-written, with mostly clear and concise explanations of ideas and mostly proper use of grammar and formatting. (2.5%)
- The project report is adequately written, with some clarity issues and inconsistent use of grammar and formatting. (2%)
- The project report is poorly written, with unclear or confusing explanations and significant errors in grammar and formatting. (1.5%)
- The project report is very poorly written, with numerous clarity and formatting errors, making it difficult to understand. (1%)

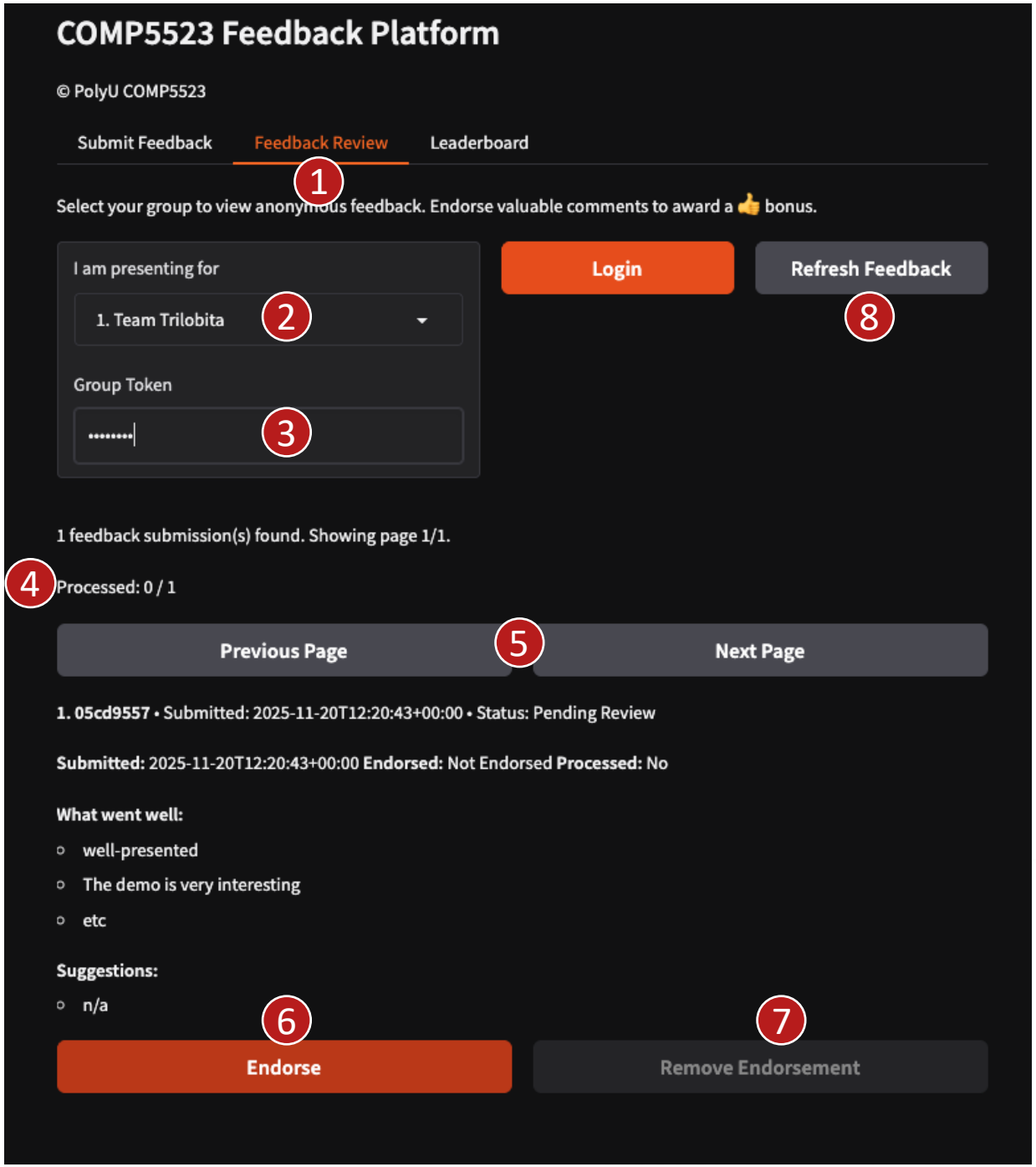


03 Online Feedback System

How to submit feedback

1. Click [Submit Feeback] at the top bar
2. Fill your name and student ID, both are required

   [Only visible to you and the teaching team; anonymous to other students]
3. Select a target group
4. Fill [What did the group do well]
5. Fill [What could be improved]
6. Click [Submit Feedback] Button
7. Status message



03 Online Feedback System

How to review feedback

1. Click [Feeback Review] at the top bar
2. Select your group
3. Input your group token [Don’t disclose it]
4. Check progress [endorsed/submitted]
5. Return to previous page or move to next page
6. Endorse feedback
7. Remove endorsement
8. Refresh feedback to show latest feedback (if have)

Best of luck
with your group project!